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Experiment 5

5. a. Implement Warshall's Algorithm to determine the transitive closure of a given directed graph represented as an adjacency matrix. The transitive closure of a graph determines all the vertices reachable from every other vertex in the graph, considering all possible paths. Warshall's Algorithm efficiently computes this closure using dynamic programming. The program should prompt the user to input the directed graph as an adjacency matrix and then apply Warshall's Algorithm to compute the transitive closure. Finally, print the resulting transitive closure matrix.

Type of Lab Activity: Exercise

Aim	Material	Methods	Answer
Given	Given	Given as Algorithm	Open

Algorithm

ALGORITHM *Warshall*($A[1..n, 1..n]$)

//Implements Warshall's algorithm for computing the transitive closure

//Input: The adjacency matrix A of a digraph with n vertices

//Output: The transitive closure of the digraph

$R^{(0)} \leftarrow A$

for $k \leftarrow 1$ **to** n **do**

for $i \leftarrow 1$ **to** n **do**

for $j \leftarrow 1$ **to** n **do**

$R^{(k)}[i, j] \leftarrow R^{(k-1)}[i, j] \text{ or } (R^{(k-1)}[i, k] \text{ and } R^{(k-1)}[k, j])$

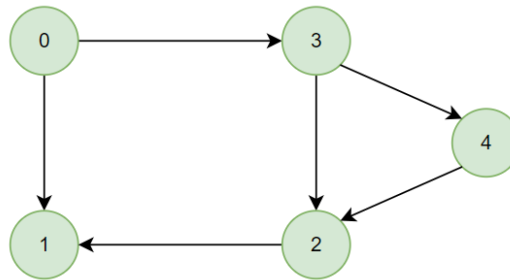
return $R^{(n)}$

Program

******(Write your answer here)******



Sample Output



Enter the number of vertices: 5

Enter the adjacency matrix (space-separated):

```
0 1 0 1 0
0 0 0 0 0
0 1 0 0 0
0 0 1 0 1
0 0 1 0 0
```

Transitive closure of the given graph is:

```
0 1 1 1 1
0 0 0 0 0
0 1 0 0 0
0 1 1 0 1
0 1 1 0 0
```